RuneScape Player Retention – Technical Practice Exercise

Level: Associate / Mid-Level Data Analyst

📌 Context

This exercise uses a fictional but realistic panel-based dataset inspired by transactional data commonly used in analytics interviews. The dataset represents player purchases observed across different cities of a classic MMORPG game, with spend of each player in the panel for 2021 and 2022.

In this Excel, you will find a sheet including 150K rows of player-level transaction across two years (2021-2022).

Your assignment is to calculate annual player retention for Prifddinas, keeping in mind that this is a panel-based, transaction-level dataset, and explain how this information could help inform player engagement and retention strategies. Retention can be defined as the percentage of players who made at least one observed purchase in Prifddinas in 2021 and were observed purchasing again in Prifddinas in 2022.

Players who were observed purchasing in Prifddinas in 2021 but did not return to Prifddinas in 2022, yet were observed spending in **other cities**, should be considered **migrated players**, and their observed purchases in other cities represent **observed migration rather than confirmed churn**.

It is important to be thoughtful about what is included in the data, as well as the **limitations inherent to panel-based datasets**, including partial observability, panel attrition, and potential data quality issues that may impact interpretation.

As with real-world panel data, this dataset intentionally contains **data quality issues and inconsistencies** that candidates are expected to identify and reason about.

*This exercise is intended for **training and educational purposes only**, not for evaluation for a real role.*

Classificado como Público